

Data Flow Diagram

Date	04 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., User, Google Gemini API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Calculate Financial Health Score).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Users, Debts, Expenses tables).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram for FinRelief AI (Level-1 DFD):

- User** → submits registration/login credentials → **Auth Process** → validates & issues JWT → **Users Data Store**.
- User** → submits debts/expenses/income → **Profile & Debt Management Process** → stores records → **Debts / Expenses / Financial Profile Data Store**.
- Dashboard Process** → reads from Debts, Expenses, Financial Profile stores → computes totals, credit utilization and charts → returns to **User**.
- AI Analysis Process** → sends financial summary → **Google Gemini API** (external entity) → receives AI-generated advice → stores in **AI Reports Data Store** → returned to **User**.
- Repayment Planner Process** → reads Debts store → applies Snowball/Avalanche logic → returns month-wise plan to **User**.
- Report Process** → reads all financial data → generates PDF using ReportLab → returned to **User** as downloadable file.